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On the Limit of the Largest Eigenvalue of the Large Dimensional Sample Covariance Matrix*

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Summary. In this paper the authors show that the largest eigenvalue of the sample covariance matrix tends to a limit under certain conditions when both the number of variables and the sample size tend to infinity. The above result is proved under the mild restriction that the fourth moment of the elements of the sample sums of squares and cross products (SP) matrix exist.

1. Introduction

Let $\{w_{ij},\ i,j=1,2,\ldots\}$ be a double array of i.i.d. random variables with mean zero and variance $\sigma^2>0$. Also, let $W_n=(w_{ij})$: $i=1,2,\ldots,p$, $j=1,2,\ldots,n$, be a matrix of order $p\times n$. Set $X_n=\frac{1}{n}W_nW_n'$, which is regarded as the sample covariance matrix with dimension p and degrees of freedom n. Suppose Y_n is another sample covariance matrix with dimension p and degrees of freedom n', independent of X_n, positive definite with probability one. Then $X_nY_n^{-1}$ is known as the multivariate F matrix with dimension p and degrees of freedom (n,n'). Suppose now that A is a $p\times p$ symmetric matrix and its eigenvalues are $\lambda_1,\lambda_2,\ldots,\lambda_p$. Define the empirical distribution of $\lambda_1,\lambda_2,\ldots,\lambda_p$, by

$$F^A(x)=\frac{1}{p}\#\{i\leqq p,\ \lambda_i\leqq x\},\tag{1.1}$$

known as the spectral distribution of the matrix A, where $\#\{\cdot\}$ denotes the number of elements of the set described by $\{\cdot\}$.

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In many situations of multivariate statistical inference, the statistics are involved with the eigenvalues of sample covariance matrices and/or those of multivariate F matrices. However, when the dimension and degrees of freedom are both large, the computation of such statistics directly from the covariance matrices or the multivariate F matrices is usually not easy. Since such statistics are usually an integral of certain functions with respect to the spectral distribution of the covariance matrices or that of the multivariate F matrices, it is of interest to find the limiting distribution of such spectral distributions and then to use the integral with respect to the limit approximating the real one. In recent years, the study in this area has received a considerable amount of attention. For some works of this area, the reader is referred to Bai, Yin and Krishnaiah (1986, 1987), Grenander and Silverstein (1977), Jonsson (1982), Wachter (1978, 1980), Yin (1986), Yin et al. (1983) and Yin and Krishnaiah (1983, 1986a, b).

Since the integrand involved is in general unbounded, for example, the determinant of the covariance matrix X_n can be written as $\det X_n = \exp\left\{p \int_0^\infty \log x \, dF^{X_n}(x)\right\}$, and the application of Helly-Bray theorem needs the boundedness of the integrand, so the limiting behavior of the extreme eigenvalues of covariance matrices and multivariate F matrices is of special interest in this area.

Now suppose $p=p(n)$ with $p/n \to y \in (0, \infty)$. $\lambda_{\max}$ denotes the largest eigenvalue of X_n. Geman (1980) proved that $\lambda_{\max} \to (1+\sqrt{y})^2 \sigma^2$ a.s. under the conditions that $E|w_{11}|^n \leqq n^{\alpha n}$, for any $n \geqq 3$ and some $\alpha > 0$. Jonsson (1983) announced the same is true when $E|w_{11}|^7 < \infty$ by using a "truncation" method. Silverstein (1984) proved the same under the condition that $E|W_{11}|^{6+\varepsilon} < \infty$. In the present paper, we shall prove the above result under the condition that $E|W_{11}|^4 < \infty$ which seems cannot be further weakened.

2. Preliminaries

The following results are needed in the sequel:

Lemma 2.1. *For any $\delta > 0$, we have*

$$\binom{k}{r}\binom{k}{l-r}\binom{2k-l}{l}\delta^{k-l} \leqq (1+\sqrt{\delta})^{2k}\binom{k}{l}\binom{2l}{2r}$$

for all $1 \leqq r \leqq l \leqq k$, where $\binom{n}{m} = n!/m!(n-m)!$.

Proof. Let

$$I(r) = \binom{k}{r}\binom{k}{l-r}\Big/\binom{2l}{2r} \quad \text{for } r = 0, 1, \ldots, l.$$

Then, we have for $r=0, 1, 2, \ldots, l-1$,

$$\frac{I(r+1)}{I(r)}=\frac{(2r+1)(k-r)}{[2(l-r)-1][k-(l-r)+1]}.$$

But

$$\begin{aligned}&(2r+1)(k-r)-[2(l-r)-1][k-(l-r)+1]\\&\quad=(2r-l+1)(2k-2l+1)\geqq 0 \quad \text{iff } 2r\geqq l-1.\end{aligned}$$

So,

$$\frac{I(r+1)}{I(r)}\geqq 1 \quad \text{iff } 2r\geqq l-1.$$

Hence $I(r)$ has its maximum at $I(0)=I(l)$, and so

$$\left(\binom{k}{r}\binom{k}{l-r}\binom{2k-l}{l}\Big/\binom{k}{l}\binom{2l}{2r}\right)\delta^{k-l}\leqq\binom{2k-l}{l}\delta^{k-l}\leqq\sum_{l=0}^{k}\binom{2k-l}{l}\delta^{k-l}$$

$$=\sum_{l=0}^{k}\binom{k+l}{2l}\delta^{l}\leqq\sum_{l=0}^{2k}\binom{2k}{l}\delta^{l/2}=(1+\sqrt{\delta})^{2k}.$$

In studying strong limit properties of random matrix, the techniques of truncation and centralization play an important role. So we shall present two lemmas concerning the truncation and centralization.

Lemma 2.2 (Truncation lemma). *Let r be a number belonging to the interval $[\frac{1}{2}, 2]$ and let $\{w_{ij},\ i, j=1, 2, \ldots\}$ be a collection of iid random variables with $Ew_{11}=0$ and $E|w_{11}|^{2/r}<\infty$. For each n, define W_n to be $p\times n$ matrix whose (i, j)-th entry is w_{ij}, where $p=p(n)$ is such that $p/n\to y\in(0, \infty)$ as $n\to\infty$. Then there exists a sequence of positive numbers $\delta=\delta_n$ such that*

(a) $\delta\to 0$ *as* $n\to\infty$

(b) *the convergence rate of δ can be slower than any preassigned rate,*

(c) $P(W_n\neq\hat{W}_n,\ i.o.)=0$

where $\hat{W}$ is the $p\times n$ matrix whose (i, j)-th entry is $\hat{w}_{ij}=w_{ij}I_{[|w_{ij}|<\delta n^r]}$ and I_A denotes the indicator of the set A.

Proof. Since $E|w_{11}|^{2/r}<\infty$, we have for any $\varepsilon>0$

$$\sum_{m=1}^{\infty}2^{2m}p(|w_{11}|\geqq\varepsilon 2^{mr})<\infty.$$

Because of the arbitrariness of ε in the above inequality, there exists a sequence of positive numbers $\varepsilon=\varepsilon_m$ such that

(1) $\varepsilon_m\to 0$, when $m\to\infty$,

(2) the rate of convergence is slower than any preassigned rate,

(3) $\sum_{m=1}^{\infty}2^{2m}P(|w_{11}|\geqq\varepsilon_m 2^{mr})<\infty.$

Define $\delta=\delta_n=2^r\varepsilon_m$ for $2^{m-1}\leqq n<2^m$. It is obvious that such a sequence of δ satisfies the requirements (a) and (b). Define $\hat{W}_n$ with this δ. Because $\frac{p}{n}\to y$, $0<y<\infty$, we have $p\leqq 2yn$ when n is large enough. Thus

$$\begin{aligned}
P(W_n\neq\hat{W}_n,\text{ i.o.})&\leqq\lim_{k\to\infty}\sum_{m=k}^{\infty}P\left(\bigcup_{2^{m-1}\leqq n<2^m}\bigcup_{i=1}^{p}\bigcup_{j=1}^{n}(|w_{ij}|\geqq\delta n^r)\right)\\
&\leqq\lim_{k\to\infty}\sum_{m=k}^{\infty}P\left(\bigcup_{2^{m-1}\leqq n<2^m}\bigcup_{i=1}^{2y2^m}\bigcup_{j=1}^{2^m}(|w_{ij}|\geqq\varepsilon_m 2^{mr})\right)\\
&=\lim_{k\to\infty}\sum_{m=k}^{\infty}P\left(\bigcup_{i=1}^{2y2^m}\bigcup_{j=1}^{2^m}(|w_{ij}|\geqq\varepsilon_m 2^{mr})\right)\\
&\leqq\lim_{k\to\infty}2y\sum_{m=k}^{\infty}2^{2m}P(|w_{11}|\geqq\varepsilon_m 2^{mr})=0
\end{aligned}$$

which completes the proof.

Lemma 2.3 (Centralization lemma). *Under the assumptions of Lemma 2.2, we have*

$$|\tilde{\lambda}_{\max}(n)-\hat{\lambda}_{\max}(n)|\to 0\quad\text{a.s.}\tag{2.1}$$

where $\tilde{\lambda}_{\max}(n)$ *and* $\hat{\lambda}_{\max}(n)$ *are the largest eigenvalues of* $\frac{1}{n^{2r}}\tilde{W}_n\tilde{W}_n'$ *and* $\frac{1}{n^{2r}}\hat{W}_n\hat{W}_n'$, *respectively, and* $\tilde{W}_n$ *is the* $p\times n$ *matrix whose* (i,j)*-th entry is* $\hat{w}_{ijn}-E\hat{w}_{ijn}$.

Proof. Denote by M_n the $p\times n$ matrix whose entries are all $E\hat{w}_{11n}$. Since $r\in[\frac{1}{2},2]$,

$$|E\hat{w}_{11n}|\leqq E|w_{11}I_{[|w_{11}|\geqq\delta n^r]}|\leqq E|w_{11}|^{2/r}n^{-2+r}\delta^{-\frac{2}{r}+1},$$

we obtain

$$\begin{aligned}
&|\tilde{\lambda}_{\max}(n)-\hat{\lambda}_{\max}(n)|\\
&\quad\leqq n^{-2r}\{2\sup_{\|a\|=1}|a'\hat{W}_nM_n'a|+\sup_{\|a\|=1}|M_na|^2\}\\
&\quad=n^{-2r}\left\{2\sup_{\sum_{i=1}^{p}a_i^2=1}\left|\sum_{i=1}^{p}a_i\right|\left|\sum_{i=1}^{p}\sum_{j=1}^{n}a_i\hat{w}_{ijn}E\hat{w}_{11n}\right|+n|E\hat{w}_{11n}|^2\sup_{\sum_{i=1}^{p}a_i^2=1}\left(\sum_{i=1}^{p}a_i\right)^2\right\}\\
&\quad\leqq cn^{-2r}\left\{n^{-1+r}\delta^{-\frac{2}{r}+1}\left(\sum_{i=1}^{p}\sum_{j=1}^{n}w_{ij}^2\right)^{1/2}+n^{-2+2r}\delta^{-\frac{4}{r}+2}\right\}\\
&\quad\leqq c\left\{n^{-1}\delta^{-\frac{2}{r}+1}\left(n^{-2r}\cdot\sum_{i=1}^{p}\sum_{j=1}^{n}w_{ij}^2\right)^{1/2}+n^{-2}\delta^{-\frac{4}{r}+2}\right\}\to 0,\quad\text{a.s.}\quad\text{if } r>1,\\
&\quad\leqq c\left\{n^{-r}\delta^{-\frac{2}{r}+1}\left(n^{-2}\sum_{i=1}^{p}\sum_{j=1}^{n}w_{ij}^2\right)^{1/2}+n^{-2}\delta^{-\frac{4}{r}+2}\right\}\to 0,\quad\text{a.s.}\quad\text{if } r\leqq 1.
\end{aligned}\tag{2.2}$$

In proving the above result, we have used the facts that

$$n^{-2r}\sum_{i=1}^{p}\sum_{j=1}^{n} w_{ij}^2 \to 0, \quad \text{a.s.} \qquad \text{if } r>1,$$

and

$$n^{-2}\sum_{i=1}^{p}\sum_{j=1}^{n} w_{ij}^2 \to E\,w_{11}^2, \quad \text{a.s.} \qquad \text{if } r\leqq 1,$$

which can be seen from Marcinkiewicz strong law of large numbers.

Remark 1. Throughout this paper, we will use Lemmas 2.2 and 2.3 with $r=\frac{1}{2}$, and the requirement (b) is specified as

$$\text{(b')} \quad \delta \log n \to \infty, \qquad \text{when } n \to \infty. \tag{2.3}$$

Remark 2. From Lemmas 2.2 and 2.3 we can easily see that

$$\lambda_{\max}(n) - \tilde{\lambda}_{\max}(n) \to 0, \quad \text{a.s.} \qquad \text{as } n \to \infty,$$

if the conditions of Theorem 1 hold, where $\lambda_{\max}(n)$ is the largest eigenvalue of $\frac{1}{n} W_n W_n'$.

3. Some Results on Graph Theory

Given a sequence $(i_1, j_1, i_2, j_2, \ldots, i_k, j_k)$, where $i_1, \ldots, i_k$ are integers in the set $\{1, \ldots, p\}$, and $j_1, \ldots, j_k$ are integers in the set $\{1, \ldots, n\}$, we define a directed multigraph as follows. We draw two parallel real lines, I-line and J-line. We plot $i_1, \ldots, i_k$ on the I-line and plot $j_1, \ldots, j_k$ on the J-line. These are vertices and they are split into two disjoint classes on the two parallel lines. So even if the two integers i_a and j_b are equal, they will not be the same vertex because i_a belongs to I-line and j_b belongs to J-line. But if $i_a = i_b$ (or $j_a = j_b$), we regard these two vertices identical. Edges of the directed bigraph will be the directed segments $\overrightarrow{i_1 j_1}, \overrightarrow{j_1 i_2}, \overrightarrow{i_2 j_2}, \overrightarrow{j_2 i_3}, \ldots, \overrightarrow{i_k j_k}, \overrightarrow{j_k i_1}$. They are $2k$ in number and they should be regarded different from each other, even in the case when two edges have the same initials and ends.

Sometimes it is convenient to denote i_a by v_{2a-1} and j_a by v_{2a}. So the vertices of the graph are $v_1, v_2, \ldots, v_{2k}$, and the edges are $v_1 v_2, v_2 v_3, \ldots, v_{2k-1} v_{2k}, v_{2k} v_1$. Notice that when we write an edge as $v_a v_{a+1}$, we always mean that v_a is the initial vertex and v_{a+1} is the end vertex, the direction of the edge is from v_a to v_{a+1}.

When two edges $v_a v_{a+1}$, $v_b v_{b+1}$ have the same vertex sets, i.e. $\{v_a, v_{a+1}\} = \{v_b, v_{b+1}\}$, we say that the two edges coincide. The reader should note that for $v_a v_{a+1}$ and $v_b v_{b+1}$ coincident, we cannot conclude that $v_a v_{a+1} = v_b v_{b+1}$, since $v_a v_{a+1} = v_b v_{b+1} \Leftrightarrow a = b$.

The graph we just constructed will be called a W-graph, if each edge coincides at least one other edge.

A W-graph will be called canonical, if $v_a \leqq \max\{v_{a-2}, v_{a-4}, \ldots\}+1$, for each $a>2$, and $v_1=1$, $v_2=1$.

In the following, we will get a bound for the number of canonical W-graphs.

In a canonical W-graph, an edge $v_{a-1}v_a$ $(a\geqq 2)$ will be called an innovation, if v_a does not occur in $v_1, v_2, \ldots, v_{a-1}$. Suppose $v_{a-1}v_a$ is an innovation. If a is odd, $v_{a-1}v_a$ is called a row innovation, and if a is even, a column innovation. Note that v_1v_2 is a column innovation according to the above definition.

In a W-graph, an edge $v_{a-1}v_a$ $(a\geqq 2)$ is said to be single up to v_b $(b\geqq a)$, if there is no other edge $v_{c-1}v_c$ with $1<c\leqq b$ such that $v_{c-1}v_c$ coincides with $v_{a-1}v_a$. An edge $v_{b-1}v_b$ $(b\geqq 3)$ will be called a T_3-edge if there is an innovation $v_{a-1}v_a$, single up to v_{b-1}, and $v_{b-1}v_b$, $v_{a-1}v_a$ coincide.

An edge will be called a T_4-edge if it is neither an innovation nor a T_3-edge.

A consecutive segment $v_av_{a+1}\ldots v_{b-1}v_b$ of the whole W-graph is called a chain.

Lemma 3.1. *Let $v_av_{a+1}\ldots v_c$ be a chain such that*

(1) *v_av_{a+1} is single up to v_c,*

(2) *v_c has been visited by $v_1v_2\ldots v_a$.*

Then the chain contains at least one T_4-edge.

Proof. When $c-a=1$, evidently $v_av_{a+1}=v_av_c$ is an edge of T_4.

We know that $v_{c-1}v_c$ must be a T_4 or a T_3 since it cannot be an innovation. If it is a T_4, the proof is completed. If $v_{c-1}v_c$ is a T_3, then there is a single innovation $v_{b-1}v_b$ coincident with $v_{c-1}v_c$ such that $b<c$.

Case 1. $b\geqq a+1$. Since $v_c=v_{b-1}$ or $v_c=v_b$, then either $v_av_{a+1}\ldots v_{b-1}$ or $v_av_{a+1}\ldots v_b$ has the properties (1), (2) and its length is shorter than $v_av_{a+1}\ldots v_c$. By induction hypothesis $v_av_{a+1}\ldots v_b$ contains a T_4-edge, but it is a part of the chain $v_av_{a+1}\ldots v_c$. So the original path contains a T_4-edge.

Case 2. $b<a+1$, i.e. $v_{b-1}v_b$ is in the path $v_1v_2\ldots v_a$, and then $v_{c-1}=(v_b$ or $v_{b-1})$ is visited by $v_1v_2\ldots v_a$. Thus $v_av_{a+1}\ldots v_{c-1}$ has the properties (1), (2). By induction, the lemma is proved.

Note. In a W-graph, the chain $v_1v_2\ldots v_{a+1}$ determines completely whether the edge v_av_{a+1} is an innovation, a T_3 or a T_4.

Lemma 3.2. *If in the chain $v_1v_2\ldots v_a$, there are s innovation edges, each of which is single up to v_a and has a vertex equal to v_a and if t is the number of noncoincident T_4 edges in $v_1v_2\ldots v_a$, then $s\leqq t+1$.*

Proof. We only need to prove the lemma for $s\geqq 2$. Let $v_{a_1-1}v_{a_1}, v_{a_2}v_{a_2+1}, \ldots, v_{a_s}v_{a_s+1}$ be all the innovation edges single up to v_a such that $a_1\leqq a_2<\ldots<a_s<a$ and $v_{a_2}=\ldots=v_{a_s}=v_a$, $v_{a_1}=v_a$ or $v_{a_1-1}=v_a$.

Consider chains $L_2=v_{a_2}v_{a_2+1}\ldots v_{a_3}$, $L_3=v_{a_3}v_{a_3+1}\ldots v_{a_4}$, $\ldots$, $L_s=v_{a_s}v_{a_s+1}\ldots v_a$. By Lemma 3.1, L_2 has a T_4-edge E_2. Let v_{b_3} be the first vertex in L_3 which belongs to L_2. Then by Lemma 3.1 $v_{a_3}v_{a_3+1}\ldots v_{b_3}$ contains an edge of T_4 and we denote it by E_3. Evidently, E_3 and E_2 are not coincident. Let v_{b_4} be the first vertex of L_4 which also belongs to $L_2\cup L_3$. Then $v_{a_4}v_{a_4+1}\ldots v_{b_4}$ has an edge of T_4, by Lemma 3.1. Let it be denoted by E_4. Evidently no two

of E_2, E_3, E_4 are coincident (see Fig. 1). Continue this procedure. finally, we get $s-1$ edges of T_4, which are not coincident with each other. So $s-1 \leqq t$, $s \leqq 1+t$.

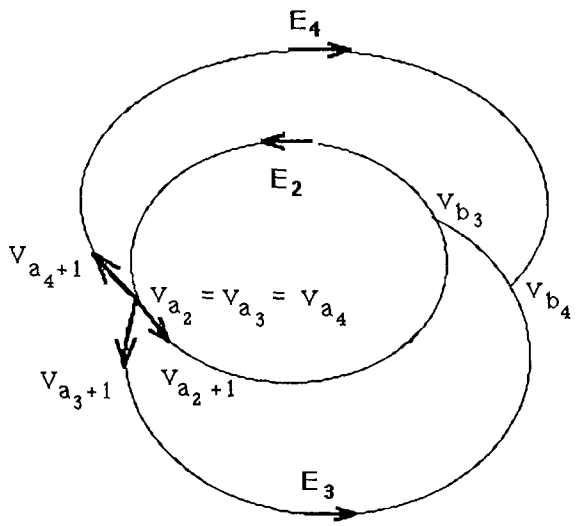


Fig. 1

A T_3 edge $v_a v_{a+1}$ is called regular if there are more than one innovations with a vertex equal to v_a and single up to v_a.

Lemma 3.3. *In any W-graph there is a mapping Φ from regular T_3 edges to T_4 edges such that for any T_4 edge E, there are at most two regular T_3 edges whose Φ image is E.*

To prove this lemma, we need the following concepts:

Definition 3.1. A chain $v_a v_{a+1} \ldots v_b$ $(b>a)$ is called a *$*$-circle*, if

(1) $v_a = v_b \notin \{v_{a+1}, \ldots, v_{b-1}\}$, $v_a = v_b$ is called the *head* of this $*$-circle,

(2) $v_a v_{a+1}$ is an innovation single up to v_b. $v_a v_{a+1}$ is called the *leader* of the $*$-circle,

(3) in $v_1 v_2 \ldots v_a$ there is at least one innovation single up to v_b and with one vertex equal to v_b, the latest among such innovations is called the *preleader* of the $*$-circle.

Definition 3.2. A $*$-circle is said to be of *first type*, if there is no T_4 edge from its preleader to its leader. Otherwise, it is said to be of *second type*.

Proof of Lemma 3.3. Suppose $v_a v_{a+1}$ is a regular T_3. Then in $v_1 v_2 \ldots v_a$, there is at least one $*$-circle with head v_a, whereas in $v_1 v_2 \ldots v_{a+1}$, one of such $*$-circles disappeared. Therefore the number of regular T_3 does not exceed the number of $*$-circles. To prove Lemma 3.3, we only need to prove the number of $*$-circles does not exceed twice the number of T_4 edges.

Let Φ^* be the mapping from $*$-circles into T_4 edges such that the Φ^* image of a first type $*$-circle is the last T_4 edge in the $*$-circle and that of a second type $*$-circle is the first T_4 in the $*$-circle. By Lemma 3.1, there is at least one T_4 edge in each $*$-circle. Hence the mapping Φ^* is well defined.

To prove the lemma, we only need to prove that there are no three $*$-circles which have a common Φ^* image. If not, then there are either two first type or two second type $*$-circles which have the same T_4 edge as their Φ^* image. Suppose they are $v_{a_1}v_{a_1+1}\ldots v_{b_1}$ and $v_{a_2}v_{a_2+1}\ldots v_{b_2}$ and $a_1<a_2$. It is evident that they must have different heads. We consider the following cases.

Case 1. Suppose that the two $*$-circles are both of the first type. We claim the $v_{b_1-1}v_{b_1}$ must be a T_4. If not, it should be a T_3, then there is an innovation $v_{c-1}v_c$ $(c<b_1)$, with one vertex equal to v_{a_1} and being single up to v_{b_1-1}. Let $v_{d-1}v_d$ be the preleader of $v_{a_1}v_{a_1+1}\ldots v_{b_1}$. Then $v_d=v_{a_1}$, since $v_{d-1}=v_{a_1}$ implies that there is at least one T_4 in the chain $v_{d-1}v_d\ldots v_{a_1}$ by Lemma 3.1, contradictory to the definition of first type $*$-circle. Therefore $d<c$, since v_d is the first appearance of the vertex v_{a_1}. Also, we have $c<a_1$ by the definition of $*$-circle and the facts that $c\leqq b_1-1$, v_c or v_{c-1} equal to v_{a_1}. By Lemma 3.1, there is at least one T_4 in the chain $v_{c-1}v_c\ldots v_{a_1}$ which is in turn contradictory to the definition of first type $*$-circle. Thus, the Φ^* image of $v_{a_1}v_{a_1+1}\ldots v_{b_1}$ is $v_{b_1-1}v_{b_1}$. By the same reason, the Φ^* image of $v_{a_2}v_{a_2+1}\ldots v_{b_2}$ is $v_{b_2-1}v_{b_2}$. Since $v_{b_1}\neq v_{b_2}$, thus $v_{b_1-1}v_{b_1}\neq v_{b_2-1}v_{b_2}$. Therefore, Case 1 is impossible.

Case 2. Suppose that the two $*$-circles are both of second type. Then there are totally three possible situations:

(1) $a_1<b_1\leqq a_2<b_2$,

(2) $a_1<a_2<b_1<b_2$,

(3) $a_1<a_2<b_2<b_1$.

In situation (1), it is evident that the Φ^* images of the two $*$-circles are different.

Suppose (2) happens. Let $v_{d-1}v_d$ be the preleader of $v_{a_2}v_{a_2+1}\ldots v_{b_2}$. If $d\leqq a_1$, then v_{a_2} $(=v_{d-1}$ or $v_d)$ has been visited (in both cases) by $v_1,\ldots,v_{a_1}$. Hence, by Lemma 3.1, there is a T_4 in the chain $v_{a_1}v_{a_1+1}\ldots v_{a_2}$. If $d>a_1$, by the definition of second type $*$-circle, there is a T_4 in the chain $v_{d-1}v_d\ldots v_{a_2}$ which is a subchain of $v_{a_1}v_{a_1+1}\ldots v_{a_2}$. For either case there is a T_4 in the chain $v_{a_1}v_{a_1+1}\ldots v_{a_2}$. Hence the Φ^* image of $v_{a_1}v_{a_1+1}\ldots v_{b_1}$ is in the chain $v_{a_1}v_{a_1+1}\ldots v_{a_2}$ while the Φ^* image of $v_{a_2}v_{a_2+1}\ldots v_{b_2}$ is behind v_{a_2}. Therefore the two $*$-circles cannot have the same Φ^* image (see Fig. 2a, b).

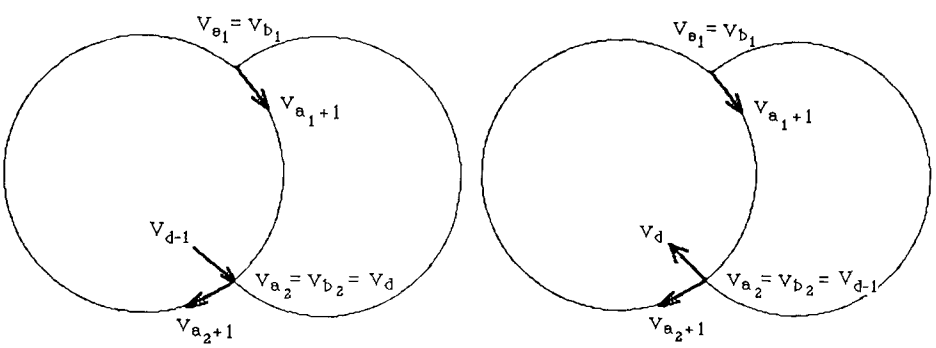


Fig. 2a, b

The argument for situation (3) is almost the same as above and omitted (see Fig. 3a, b).

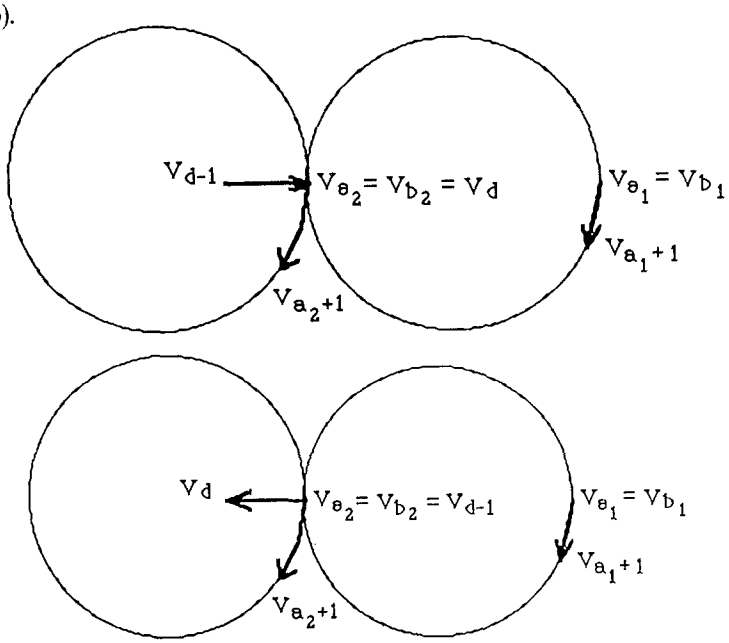


Fig. 3a, b

Summing the discussion for the two cases, the proof of the lemma is complete.

4. Limit of the Largest Eigenvalue

We now prove the main result of our paper.

Theorem 3.1. *Let* $\{w_{ij}: i,j=1,2,\ldots\}$ *be an infinite matrix of iid random variables,* $Ew_{11}=0$ *and* $Ew_{11}^4<\infty$. *If* $\lambda_{\max}(n)$ *denotes the largest eigenvalue of the matrix* $\frac{1}{n}W_nW_n'$, *here* W_n *denotes the* $p\times n$ *random matrix* $[w_{ij}: i=1,\ldots,p;\ j=1,\ldots,n]$, *then*

$$\lim_{n\to\infty}\lambda_{\max}(n)=(1+\sqrt{y})^2\,Ew_{11}^2 \quad \text{a.s.}$$

as $n\to\infty$, $p\to\infty$ *and* $p/n\to y>0$.

Proof. Without loss of generality, we assume that $Ew_{11}^2=1$. By the result of Jonsson (1982) or Wachter (1978), we have $\underline{\lim}_{n\to\infty}\lambda_{\max}(n)\geqq(1+\sqrt{y})^2$ a.s. Hence we only have to prove that $\overline{\lim}\,\lambda_{\max}(n)\leqq(1+\sqrt{y})^2$ a.s. But, according to Remark 2 of Sect. 2, it is sufficient to show that

$$\overline{\lim}\,\tilde{\lambda}_{\max}(n)\leqq(1+\sqrt{y})^2 \quad \text{a.s.} \tag{4.1}$$

In other words, we assume in the sequel that

(1) w_{ij}, $i=1,2,\ldots,p$, $j=1,2,\ldots,n$ are i.i.d. for each n,

(2) $|w_{ij}|<\delta\sqrt{n}$,

(3) $Ew_{ij}=0$,

(4) $Ew_{ij}^2\leqq 1$, and $E_{ij}^2\to 1$ as $n\to\infty$,

(5) $E|w_{ij}|^l\leqq(\delta\sqrt{n})^{l-1}$, for $l\geqq 2$,

(6) $E|w_{ij}|^l\leqq c(\delta\sqrt{n})^{l-3}$, for $l\geqq 3$.

Now, choose $z>(1+\sqrt{y})^2$ arbitrarily. We proceed to prove (4.1) by showing that

$$\sum_{n=1}^{\infty} E\left(\frac{\lambda_{\max}(n)}{z}\right)^k<\infty, \tag{4.2}$$

where $k=k_n$ satisfies

$$k_n/\log n\to\infty, \tag{4.3}$$

$$\delta^{1/6}k_n/\log n\to 0. \tag{4.4}$$

We have

$$E[\lambda_{\max}(n)]^k\leqq E\,\mathrm{tr}\left(\frac{1}{n}WW^T\right)^k=n^{-k}E\,\mathrm{tr}(WW^T)^k$$
$$=n^{-k}\sum E\,w_{i_1j_1}w_{i_2j_1}w_{i_2j_2}\cdots w_{i_kj_k}w_{i_1j_k}.$$

Here the summation is taken in such a way that $i_1,\ldots,i_k$ run over all integers in $\{1,\ldots,p\}$, and $j_1,\ldots,j_k$ run over all integers in $\{1,\ldots,n\}$. For given $i_1,\ldots,i_k$, $j_1\ldots j_k$, construct a graph as described in Sect. 3. If a graph is not a W-graph, then it corresponds to a term equal to zero. Hence, the above sum can be split in the following form:

$$E\,\mathrm{tr}\left(\frac{1}{n}WW^T\right)^k=n^{-k}\sum{}'\sum{}''\sum{}'''E\,w_{i_1j_1}w_{i_2j_1}w_{i_2j_2}\cdots w_{i_kj_k}w_{i_1j_k}.$$

Here

$\sum'$-summation for different arrangements of four different types of edges at the $2k$ different positions.

$\sum''$-summation for different canonical graphs Γ with given arrangement of the four types for $2k$ positions.

$\sum'''$-summation of $Ew_{i_1j_1}w_{i_2j_1}\cdots w_{i_1j_k}$ for which the graph is isomorphic to the given canonical graph.

Let r denote the number of row innovations, l denote the number of T_3 edges. Then there are $l-r$ column innovations and $(2k-2l)$ T_4 edges and so $\sum'$ is bounded by $\sum_{l=1}^{k}\sum_{r=1}^{l}\binom{k}{r}\binom{k}{l-r}\binom{2k-l}{l}$. Since every row innovation leads to a free i-index and every column innovation leads to a free j-index except the first column innovation v_1v_2 which leads to an i-index and a j one. We know that $\sum'''$ is bounded by $p^{r+1}n^{l-r}$.

To bound $\sum''$, let t denote the number of noncoincident T_4 edges. By definition, each innovation in a canonical W-graph is uniquely determined by the chain before it and each irregular T_3 edge is so done as innovations. If $t=0$,

i.e. $l=k$, then by Lemma 3.3, there are no T_4 edge and regular T_3 edge, so that $\sum''$ is only one summand. For $t\geqq 1$, since each T_4 edge in a canonical W-graph must be one of the k^2 elements in the $k\times k$ matrix (w_{ij}) $i\leqq k$, $j\leqq k$, all the possibilities that $2k-2l$ T_4 edges may take are less than $\binom{k^2}{t}t^{2k-2l}$. By Lemmas 3.2 and 3.3, all the possibilities that all the regular T_3 edges take is not more than $(t+1)^{4k-4l}$. Hence $\sum''$ is bounded $\binom{k^2}{t}t^{2k-2l}(t+1)^{4k-4l}$ $\leqq k^{2t}(t+1)^{6k-6l}$.

Finally, we bound the expectation $E w_{i_1j_1}w_{i_2j_1}\cdots w_{i_kj_k}w_{i_1j_k}$. If $t=0$, each expectation is $(E w_{11}^2)^k\leqq 1$. For $t\geqq 1$, let μ denote the number of innovations which coincide with at least one T_4 edge and let n_i denote the number of T_4 edges which coincide with the i-th such innovation, $i=1,2,\ldots,\mu$, respectively.

Let m_j be the number of T_4 edges which coincide with each other but not with any innovation, $j=1,2,\ldots,t-\mu$. Then we have

$$E w_{i_1j_1}w_{i_2j_1}\cdots w_{i_kj_k}w_{i_1j_k}=(E w_{11}^2)^{l-\mu}\prod_{i=1}^{\mu}(E w_{11}^{n_i+2})\prod_{j=1}^{t-\mu}E w_{11}^{m_j}$$

where $2(l-\mu)+\sum_{i=1}^{\mu}(n_i+2)+\sum_{j=1}^{t-\mu}m_i=2k$ and $\mu\leqq t$. By (4), (5) and (6) we have

$$\begin{aligned}&|E w_{i_1j_1}w_{i_2j_1}\cdots w_{i_kj_k}w_{i_1j_k}|\\&\leqq c^{\mu}(\delta\sqrt{n})^{\sum_{i=1}^{\mu}(n_i-1)+\sum_{j=1}^{t-\mu}(m_j-1)}=c^{\mu}(\delta\sqrt{n})^{2k-2l-t}\leqq k^t(\delta\sqrt{n})^{2k-2l-t}\end{aligned}$$

when k is large enough. By the above argument, we obtain

$$\begin{aligned}&E\,\mathrm{tr}\left(\frac{1}{n}W_nW_n^T\right)^k\\&\leqq n^{-k}\sum_{l=1}^{k}\sum_{r=1}^{l}\binom{k}{r}\binom{k}{l-r}\binom{2k-l}{l}p^{r+1}n^{l-r}\left[\sum_{t=0}^{2k-2l}k^{2t}(t+1)^{6k-6l}k^t(\delta\sqrt{n})^{2k-2l-t}\right]\\&\leqq p\sum_{l=1}^{k}\sum_{r=1}^{l}\binom{k}{r}\binom{k}{l-r}\binom{2k-l}{l}\left(\frac{p}{n}\right)^r\left[\sum_{t=0}^{2k-2l}k^{3t}(t+1)^{6k-6l}(\delta\sqrt{n})^{-t}\right]\delta^{2(k-l)}.\end{aligned}$$

Using the elementary inequality $a^{-t}(t+1)^b\leqq a\left(\frac{b}{\log a}\right)^b$ for $a>1$, $b>0$, $t>0$, we have

$$\begin{aligned}\sum_{t=0}^{2k-2l}k^{3t}(t+1)^{6k-6l}(\delta\sqrt{n})^{-t}&\leqq 2k\left(\frac{\delta\sqrt{n}}{k^3}\right)\left(\frac{6k-6l}{\log\frac{\delta\sqrt{n}}{k^3}}\right)^{6k-6l}\\&\leqq 2n\left(\frac{6k}{\frac{1}{2}\log n+\log\delta-3\log k}\right)^{6(k-l)}\\&\leqq 2n\left(\frac{18k}{\log n}\right)^{6(k-l)}\qquad\text{when } n \text{ is large enough.}\end{aligned}$$

Using Lemma 2.1, we get for all large n

$$
\begin{aligned}
&E\,\mathrm{tr}\left(\frac{1}{n}W_n W_n^T\right)^k \\
&\leqq 2np\sum_{l=1}^{k}\sum_{r=1}^{l}(1+\sqrt{\delta})^{2k}\binom{k}{l}\binom{2l}{2r}\left(\frac{p}{n}\right)^r\left[\frac{18\delta^{1/k}k}{\log n}\right]^{6(k-l)} \\
&\leqq 2np(1+\sqrt{\delta})^{2k}\left[(1+\sqrt{p/n})^2+\left[\frac{18\delta^{1/6}k}{\log n}\right]^6\right]^k \\
&=\left[(2np)^{1/k}(1+\sqrt{\delta})^2\left((1+\sqrt{p/n})^2+\left[\frac{18\delta^{1/6}k}{\log n}\right]^6\right)\right]^k \\
&\leqq \eta^k,
\end{aligned}
$$

where η is a constant satisfying $(1+\sqrt{y})^2<\eta<z$. Here the last inequality follows from the following facts:

(a) $(2np)^{1/k}\to 1$, because $k/\log n\to\infty$ and $p/n\to y\in(0,\infty)$,
(b) $(1+\sqrt{\delta})^2\to 1$, because $\delta\to 0$,
(c) $(1+\sqrt{p/n})^2\to(1+\sqrt{y})^2$, because $p/n\to y\in(0,\infty)$,
(d) $\left(\frac{\delta^{1/6}k}{\log n}\right)^6\to 0$, because $\delta^{1/6}k/\log n\to 0$.

This leads to (4.2) since $k/\log n\to\infty$ and the proof is thus complete.

Remark. We have conjectured that the condition of finite 4th moment of the entries of W_n is necessary for almost sure limit of the largest eigenvalue of W_n. This was proved in Bai et al. (1986). The result is as following. If $EW_{11}^4=\infty$, then $\limsup_{n\to\infty}\lambda_{\max}=\infty$, a.s.

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